

CIVL4340 Assignment 1

Wind climate analysis

Overview

You have been contracted by the Australian government to determine appropriate design wind speeds for use in the construction of new high-importance structures they aim to build. The structures will be located near to the weather station you were assigned and analysed during the first two tutorial sessions. You must determine the design wind speeds based on the long-term wind records from your Automatic Weather Stations (AWS) and compare these with what is prescribed in AS/NZS1170.2 for your location.

Given the importance of the proposed structure, the government has requested that the wind speeds used for ultimate limit design calculations should have an exceedance probability of less than 5% for a design life of 75 years. Serviceability design calculations can be based on a wind speed with an annual probability of exceedance of 5%.

Considering the above points, you are required to determine serviceability and ultimate limit state design wind speeds for a build site, which is expected to sit in flat terrain classed as **terrain category 3**. You should assume that the building to be constructed has a reference height is **20 m**.

You can choose to use **one** of the following two approaches to estimate your design wind speeds, 1) Peak over Threshold (PoT) method (using a GPD fit), or 2) PoT method with a Type III fit and shape factor of $k = 0.1$ (as done in you Week 3 tutorial – see instructions therein). Prior to fitting any statistical distributions to your data, you should standardize your gust readings to equivalent Terrain Category 2, $z = 10$ m, 0.2 second gust values. You should also divide your data into gusts generated by different storm types (e.g. synoptic scale, thunderstorm, tropical cyclone, we will discuss ways of doing this during tutorials). Once you fit your distribution to each resulting gust data set, you will then need to combine results for each individual sub-set before identifying your final design wind speeds.

You are also required to write a short overview of the wind events that influence the extreme wind climate in the region your structure is to be built. This should include some discussion of the type of storms that generate extreme winds as well as any other interesting information drawn from your data sets (e.g. typical time of year/day for severe winds).

Dataset

You should again use the wind data assigned to you during the tutorial in Week 1. The latitude and longitude of your site was provided to you but can also be found in the station metadata files available on the [BoM website](#). This location information can be used to determine any

topographic or terrain features that may have influenced the acquired data at your weather station.

Submission

You are required to submit a short report summarising your analysis and results. As a guide, your report should include,

- A 1-page executive summary providing the requested design wind speeds. This should include all the tables listed on page 3 of this information sheet.
- A brief introduction explaining what the purpose of your report is and providing a brief review of extreme wind events in your region. Please write this as though the assignment is an actual submission to a structural engineer.
- A brief description of your observation site (i.e. the weather station location), which could include map/s and a discussion of any local site factors that may influence your analysis. Please include any adjustments you have made to your data based on these factors in this section (you can use the same tables as in the Executive Summary if you like).
- A methodology section that outlines what quality control and analysis techniques you have used.
- A results section that provides analysis results for your station. Within this section you should plot a return period/ARI curve (don't forget to include your observations) as well as providing the specific design wind speeds in tabular form. Remember to show results for each of your storm types as well as the combined curve). A short discussion of your results is expected.

Your submission is due at **4 pm** on **Thursday April 2**. Please submit your assignment through the link provided in the Assignment 1 folder on Blackboard. Please note that this is an **individual** assignment, where every student must submit their own report. Your report should be no more than approximately **6 pages** long (you can add appendices at the end if you like).

Assignment 1 is **20%** of your total mark for the course. The course lecturer will mark the assignment based on the accuracy of your results, quality of figures/tables and overall clarity of your report.

Additional notes

1. You can use the Elevation capability in Google Earth (desktop version) to determine the topographic features near the weather station site.
2. Metadata for the Automatic Weather Stations can be found [here](#).
3. Wind speed adjustments to account for instrument type can be made using the information provided on Blackboard.
4. Adjustments for terrain and topography should be made using the concepts and methods embodied in similar adjustments in AS/NZS1170.2. An example is provided on Blackboard.

Tables for Executive Summary

Recurrence intervals

Ultimate Limit State (ULS)	
Serviceability Limit State (SLS)	

Standardisation multipliers at AWS site

	N	NE	E	SE	S	SW	W	NW
M_{z,cat}								
M_t								

Multipliers for your build site

M_{z,cat}	
M_t	

Standardised regional design wind speeds (i.e. z = 10 m, terrain category 2, flat ground)

	EVA analysis	AS/NZS1170.2
ULS		
SLS		

Design wind speeds at your build site (i.e. z = 20 m, terrain category 3, flat ground)

	EVA analysis	AS/NZS1170.2
ULS		
SLS		